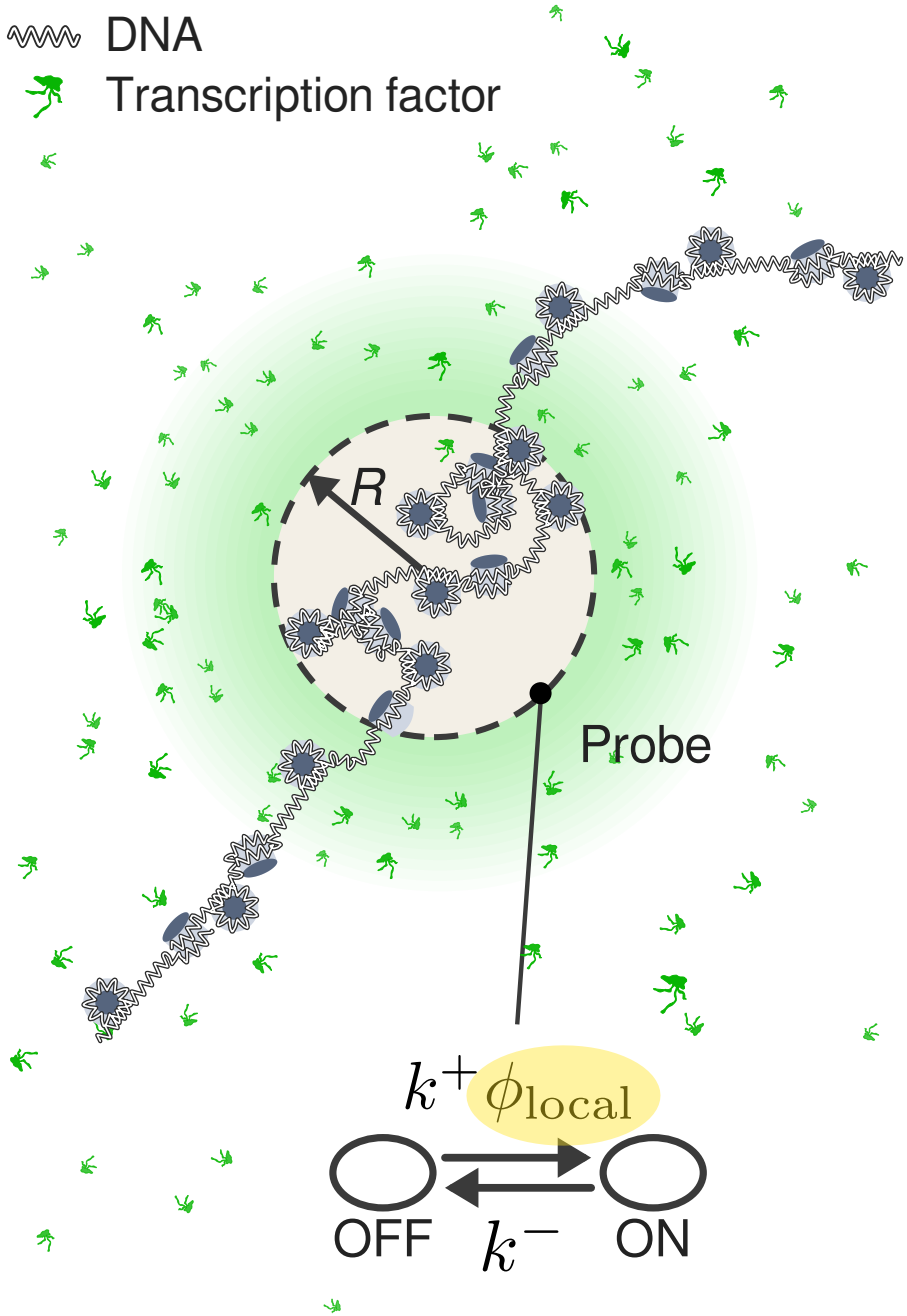


Flory-Huggins potential

$$f(\phi) = \frac{\phi}{\nu}\ln\phi + (1-\phi)\ln(1-\phi) - \chi\phi^2 - \mu\phi$$

Cahn-Hilliard total energy

$$F = \int_B \left[\frac{\kappa}{2}(\nabla\phi)^2 + f(\phi)\right] d^3x - \int_S h_0\phi(\mathbf{x})dS$$

ϕ TF volume fraction
ν TF / solvent volume ratio
χ TF-TF attraction
μ chemical potential
h_0 surface attraction
κ Interface stiffness